

Assignment 6

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Objective

The goal of this assignment is to analyze which regions of an image influence a neural network's classification decision and how adversarial perturbations affect the model's attention. We use Grad-CAM and Integrated Gradients (IG) to visualize important areas in both original and adversarial images, and compare the impact of using different layers (softmax vs preceding fully connected layer) for gradient computation.

Methodology

- **Model and Data:**
We use the pretrained VGG16 model on ImageNet. A sample image of an African elephant is loaded and preprocessed.
- **Adversarial Example Creation:**
Using the Fast Gradient Sign Method (FGSM), an adversarial example is generated by perturbing the original image with the gradient of the loss with respect to the input, scaled by 0.15.
- **Attention Visualization:**
 - *Grad-CAM*: Computes gradients of the predicted class score with respect to the activations of the last convolutional layer (`block5_conv3`), generating a spatial heatmap of important regions.
 - *Integrated Gradients*: Estimates pixel-wise importance by integrating gradients along a path from a baseline (all-zero image) to the input image.

- **Layer Investigation:**

Integrated Gradients are calculated twice: once using the softmax output layer and once using the preceding fully connected layer (**fc2**) to observe differences in attribution.

- **Comparison and Quantitative Analysis:**

Correlation coefficients between attention maps of original and adversarial images are computed for both Grad-CAM and IG, measuring similarity in model focus.

Results

Metric	Correlation (Original vs Adversarial)
Grad-CAM	0.987
Integrated Gradients	0.958
IG Softmax vs FC2	1.000

Table 1: Correlation coefficients between original and adversarial attention maps

- The **Grad-CAM heatmaps** show a very high correlation (0.987), indicating that spatial regions of importance remain mostly consistent despite adversarial perturbations.
- The **Integrated Gradients maps** also show high similarity (0.958), though slightly less than Grad-CAM, suggesting that pixel-level attributions shift subtly under adversarial influence.
- Comparing IG computed from the **softmax output** and from the **previous FC2 layer** reveals perfect correlation (1.000), demonstrating that both layers provide consistent gradient information for attribution.

The model’s top predicted class changes due to adversarial noise, but the core spatial focus in convolutional features remains largely stable.

Observations and Discussion

- Adversarial examples shift model predictions but do not drastically alter key spatial attention regions at the convolutional layer level, as shown by high Grad-CAM correlation.

- Integrated Gradients provide fine-grained pixel importance, which is slightly more sensitive to adversarial perturbations than Grad-CAM.
- Grad-CAM works best when computed from convolutional layers rather than softmax outputs, since convolutional activations preserve spatial information necessary for meaningful heatmaps.
- Integrated Gradients are robust to layer choice, yielding consistent attribution when computed from softmax or the previous fully connected layer.
- A TensorFlow warning was observed during IG calculation from the FC2 layer related to input structure; however, it did not affect gradient computation or the validity of results.

Conclusion

This assignment demonstrates how interpretability methods like Grad-CAM and Integrated Gradients can reveal the model's decision-making focus and how adversarial perturbations influence it. While adversarial noise can change the model's predicted class, the important spatial features identified by Grad-CAM remain similar, suggesting the adversarial effect is subtle and distributed rather than causing complete attention shifts.

Furthermore, the comparison of attribution layers for IG confirms that both softmax and preceding layers yield consistent explanations, providing flexibility in interpretability analysis.

Visualizations



Figure 1: Left: Original Image. Right: Adversarial Image

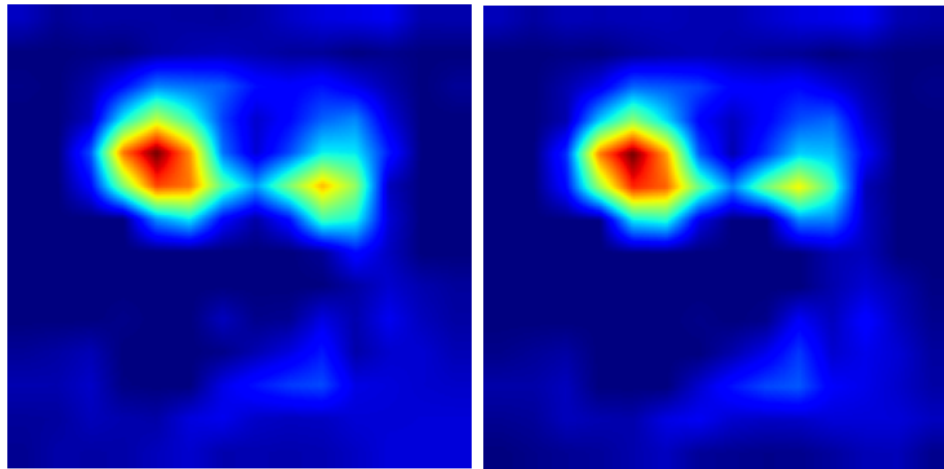


Figure 2: Grad-CAM Heatmaps for Original (left) and Adversarial (right) Images

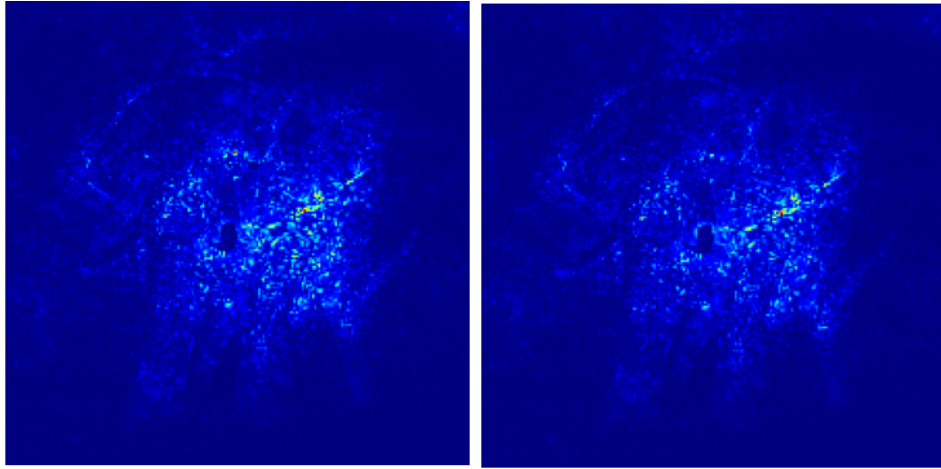


Figure 3: Integrated Gradients Heatmaps for Original (left) and Adversarial (right) Images

Source: Google Colab